BharatVoice AI – Project README

Project Title:
BharatVoice AI – Voice-to-Voice & Multilingual Learning Assistant

Student Project Overview
This project is a fully interactive AI-powered language-learning and communication system inspired by Duolingo, powered with real-time voice recognition, AI conversation, translation, and text-to-speech capabilities.

The goal of the project is to build a beginner - friendly yet feature - rich AI web application that works on both **mobile and laptop**, with a clean and colourful modern UI.

Key Features

1. Voice Tutor (Voice ↔ Text ↔ AI ↔ Voice)
- User speaks into microphone.
- Speech is converted to text.
- Text is sent to AI (via OpenRouter LLMs).
- AI generates a meaningful response.
- Browser speaks the response back to user.

2. Chat Playground (Text ↔ AI)
- Users can type messages.
- AI responds naturally and intelligently.
- Users can listen to AI replies using TTS.

3. Translator (English ↔ Indian Languages)
Supports:
- English
- Hindi
- Tamil
- Kannada
- Telugu

Translation works both ways and includes voice playback.

Technologies Used

Frontend:
- **HTML5**
- **CSS3** (gradient UI, responsive, card-based design)
- **JavaScript**
- **Web Speech API**
- Speech-to-Text
- Text-to-Speech

AI & Backend:
- **OpenRouter LLM API**

- Model used: *google/gemma-7b-it* (can be upgraded)
- Optional free fallback translator: *MyMemory API*

How It Works

Speech Recognition
Uses browser's built-in speech recognition (Chrome recommended).

AI Communication
AI replies are generated by sending prompts to OpenRouter API using:
```
https://openrouter.ai/api/v1/chat/completions
```

Text to Speech
AI replies are spoken aloud with:
```
speechSynthesis.speak()
```

Translation Engine
Free MyMemory translation API is used for demo purposes.

Project File
Your main working file:
```
index.html
```
This file contains the entire UI + logic in one place for easy running.

How to Run the Project
1. Save the file `index.html`.
2. Open it in **Google Chrome** on laptop or phone.
3. Allow microphone access.
4. Choose any tool:
- Voice Tutor
- Translator
- Chat Playground
5. Begin speaking or typing.

Deployment Options
You can share this as a real website by uploading `index.html` to:
- **Netlify Drop** (fastest)
- **GitHub Pages** (free forever)

- **Vercel** (professional hosting)

Limitations & Future Improvements
- API key must be protected using backend proxy.
- Better translation quality using LLMs could be added.
- Add progress tracking and more gamification.
- Add user authentication with local storage or Firebase.

Conclusion
This project demonstrates a complete, student - friendly implementation of real-time AI voice interaction, translation, and multilingual tutoring. The UI is modern, colorful, and responsive, designed to resemble professional apps like Duolingo.

The system is easy to extend, deploy, and present in academic or competition environments.

Project Completed By:
Student Name: ______________________
Class / Course: ____________________
Date: ______________________________